

Demo 1 Setup and Configuration

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Hardware/Software Requirements

The following instructions are presumed to be run on an Ubuntu Linux OS installed natively (no in a VM) on a computer with an NVIDIA dedicated graphics card. The following represents the software/hardware used to run Demo 1. Other configurations may work as well.

- Ubuntu Linux 20.04 LTS
- Kernel version 5.13.0-39-generic
- NVIDIA Drivers version 460
- Intel Core i9 8 core
- 64GB Ram
- 1TB Disk capacity

Building and Installing the CARMA-CARLA Integration Tool

Setup CARMA Platform

Instructions for installing the bas CARMA Platform can be found on the CARMA Platform confluence page - <https://usdot-carma.atlassian.net/wiki/spaces/CRMPLT/pages/486178827/Development+Environment+Setup> . Follow the instructions as shown to configure a development environment. Depending on the physical setup, it is possible to skip the configuration of a VM (if installing on a native Linux machine) and the configuration of VSCode (if no further development is needed) though the information contained in both steps can be useful for understanding how CARMA Platform works and is configured.

It is presumed that the CARMA platform is installed in the directory referenced in the CARMA Platform instructions, e.g. \$HOME/carma_ws.

Ensure that CARMA Platform is able to run by following some or all of the instructions on the last page of the setup instructions - <https://usdot-carma.atlassian.net/wiki/spaces/CRMPLT/pages/491651073/Using+a+CARMA+Platform+development+environment> . The commands to start "carma start all" and configure "carma config edit" are particularly important.

Download the TFHRC Map

The physical location for Demo 1 is the campus of the FHWA Turner Fairbanks Laboratory in McLean, VA. A map of the facility grounds has been developed in the CARLA Simulator.

The map and CARLA backend can be obtain in a single download.

- Ubuntu Version: <https://www.dropbox.com/sh/ga3viv4l2wxd6n9/AAARGbALNcYkGyYakNKHZzMca?dl=0>
- Windows Version: https://www.dropbox.com/sh/1hrgggo95oe375y/AAD_CWzYiU2bfRHfwsfFaAc7a?dl=0

The Ubuntu version is recommended and will be used/referenced but the Windows version can be made to work as well. To ensure the map has downloaded and is functional, unzip the file in \$HOME/carla and navigate to \$HOME/carla/CARLA_0.9.10_TFHRC_Ubuntu_20220301/LinuxNoEditor and run "./CarlaUE4.sh".

```
cd $HOME/carla/CARLA_0.9.10_TFHRC_Ubuntu_20220301/LinuxNoEditor && ./CarlaUE4.sh
```

An overview map of TFHRC should open and be navigable with game-like controls (left click and move mouse to move the camera, w,s,a,d keys to pan).

Once the map is installed, it is recommended to test it by instantiating a vehicle and driving around the grounds. CARLA provides several scripts to test this located in the "\$HOME/carla/CARLA_0.9.10_TFHRC_Ubuntu_20220301/LinuxNoEditor/PythonAPI/examples" directory. Run, manual_control.py for example with the command "python3 manual_control.py" in the examples directory.

```
cd $HOME/carla/CARLA_0.9.10_TFHRC_Ubuntu_20220301/LinuxNoEditor  
/PythonAPI/examples && python3 manual_control.py
```

When first running the example script several python addons may need to be installed. Add the CARLA python library to the python path with the following command:

```
PYTHONPATH=$PYTHONPATH;$HOME/carla/CARLA_0.9.10_TFHRC_Ubuntu_20220301  
/LinuxNoEditor/PythonAPI/carla/dist/carla-0.9.10-py3.8-linux-x86_64.egg
```

Other packages, such as numpy and pygame may need to be installed as well with the following commands:

```
pip3 install numpy pygame
```

Once you are able to drive a vehicle around the TFHRC map, move onto the next step of installing and configuring the carma-carla-integration tool.

Setup CARMA Simulation

The carla-carla-integration tool is part of the CARMA Simulation (<https://github.com/usdot-fhwa-stol/carma-simulation.git>) package and managed from a different CARMA repository than the CARMA Platform repository cloned previously in this setup. To start, clone the CARMA Simulation repository and checkout the release/carma-xil-1.0 branch.

```
mkdir $HOME/carma_sim && cd carma_sim  
git clone https://github.com/usdot-fhwa-stol/carma-simulation.git  
cd carma-simulation  
git checkout release/carma-xil-1.0
```

Navigate to the carla-carla-integration directory and build the carla-carla docker image by running the build image script.

```
cd $HOME/carma_sim/carma-simulation/carla-carla-integration && ./build-  
image.sh
```

CARMA Simulation will also require a specific CARMA configuration. The CARMA Config repository was automatically cloned with the rest of the CARMA Platform configuration. That repository will be located in \$HOME/carma_ws/src/carma_config and can be used. Alternatively it can be cloned separately into a new directory on its own, either option is acceptable. Regardless of the approach taken, the repository will need to be updated to the Update/carla_integration branch.

To use the branch previously cloned:

```
cd $HOME/carma_ws/src/carma-config  
git checkout Update/carla_integration
```

or to use a fresh carma_config:

```
mkdir $HOME/carma_config && cd $HOME/carma_config
git clone https://github.com/usdot-fhwa-stol/carma-config.git && cd
carma-config
git checkout Update/carla_integration
```

After checking out the Update/carla_integration branch a new carma-config will need to be generated by running the build_image script in the carla_integration folder.

```
cd carla_integration && ./build_image.sh
```

Once the image is built, set the carma-config image with the tagged version that results from the build. Note in the code snippet below, replace [tag] with the image tag associated with the previously run build_image command.

```
carma config set usdotfhwastol/carma-config:[tag]
```

Setup CARMA Platform with TFHRC Route

In order to test the carma-carla simulation, first a route for CARMA is needed to direct the vehicle where to travel. Two tools are required, one to convert the CARLA map from the opendrive format to lanlet format, and a second to manipulate the lanelet format to output the CARMA required route format.

First download the opendrive2lanelet tool from the CARMA github repository at <https://github.com/usdot-fhwa-stol/opendrive2lanelet>. Instructions are also available via the README file, or follow the directions below.

```
cd $HOME
git clone https://github.com/usdot-fhwa-stol/opendrive2lanelet.git
cd opendrive2lanelet
docker build -t opendrive2lanelet2convertor .
```

Next, the OpenDrive formatted map is available in the CARLA installation directory. Find it a copy it to a temporary location.

```
mkdir $HOME/openDriveMap && cd $HOME/openDriveMap
cp $HOME/carla/CARLA_0.9.10_TFHRC_Ubuntu_20220301/LinuxNoEditor/CarlaUE4
/Content/Carla/Maps/OpenDrive/TFHRC.xodr .
docker run --rm -it -v $HOME/openDriveMap:/root/opendrive2lanelet/map
opendrive2lanelet2convertor
```

When the process is finished, a file name TFHRC.osm should be created in the openDriveMap folder that was created in the previous step.

Copy the TFHRC.osm file to /opt/carma/maps/ and point the vector_map.osm link to the newly created file.

A copy of the map created for Voices Demo 1 is also here:



TFHRC.osm

```
cd /opt/carma/maps
cp $HOME/openDriveMap/TFHRC.osm .
ln -vfns /opt/carma/maps/TFHRC.osm vector_map.osm
```

Next follow the steps outlined in the instructions found in the Creating a New Route File CARMA Platform confluence page: <https://usdot-carma.atlassian.net/wiki/spaces/CRMPLT/pages/1716060161/Creating+a+New+Route+File?preview=%2F1716060161%2F1716060189%2Fimage-20210514-135105.png> . Note: Step 4 may not be required.

The resulting csv file(s) should be placed in the /opt/carma/routes folder.

Example route files for Voices Demo 1 are here:



VOICES_Demo1_v..._3_5_route.csv



VOICES_Demo1_veh1_route.csv

Running the CARMA-CARLA Simulation

The CARMA-CARLA simulation should now be ready to run. To run the simulation, three elements need to be started in a specific order. The CARMA Platform, then the CARLA simulation and finally the carma-carla-integration tool

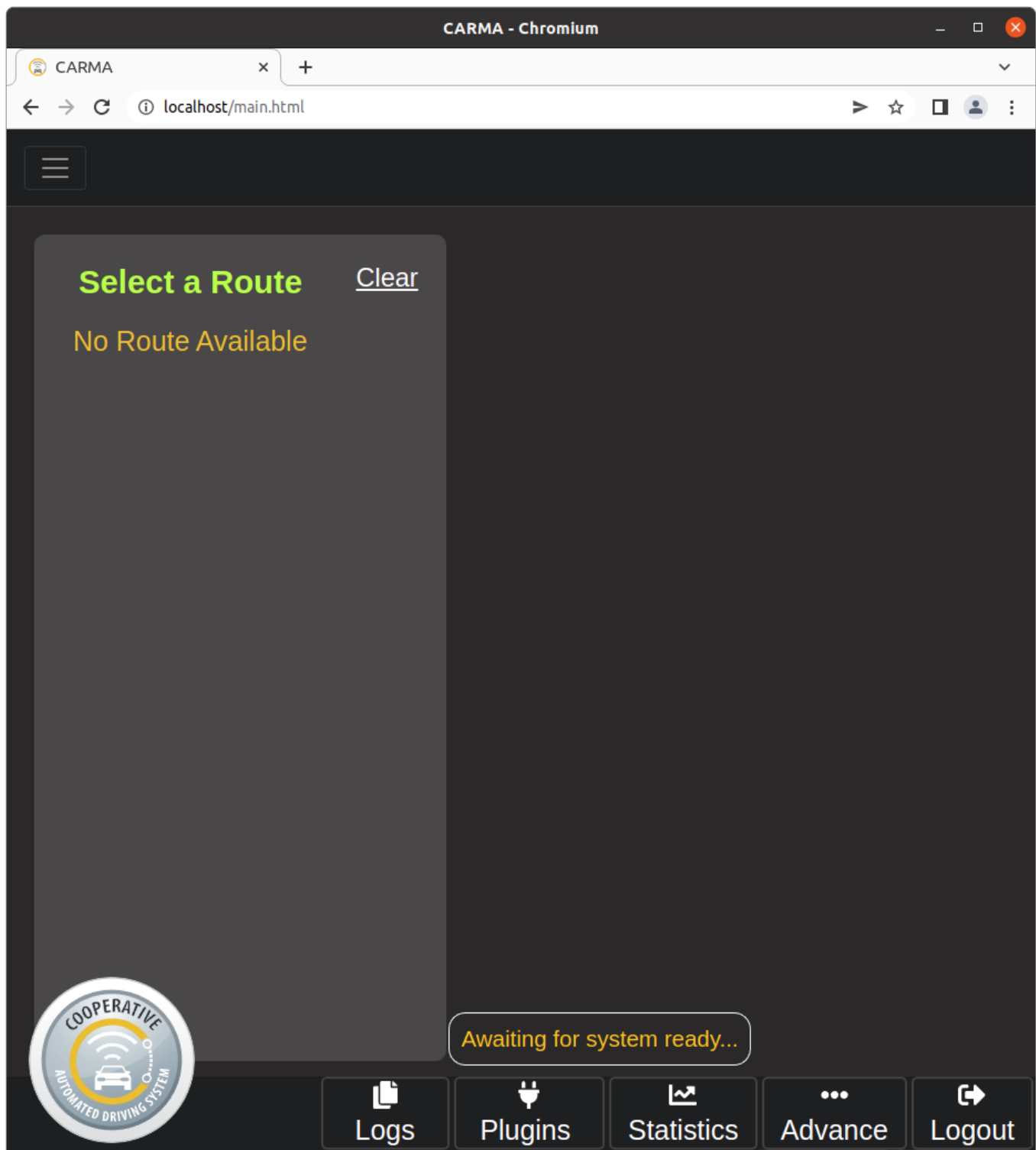
Prepare the carma-carla-integration tool by starting the docker image and running two commands to set the path and some environmental variables.

```
cd $HOME/carma_sim/carma-simulation/carla-carma-integration && ./run.sh  
export PYTHONPATH=$PYTHONPATH:/home/PythonAPI/carla-0.9.10-py2.7-linux-  
x86_64.egg && source /home/carla_carma_ws/devel/setup.bash
```

Leave that terminal open and in another terminal, start carma

```
carma start all
```

Open a chromium browser to localhost and you should see the CARMA Platform Web UI. Turn Debug Mod on in the lower right corner and click on "Start Platform"



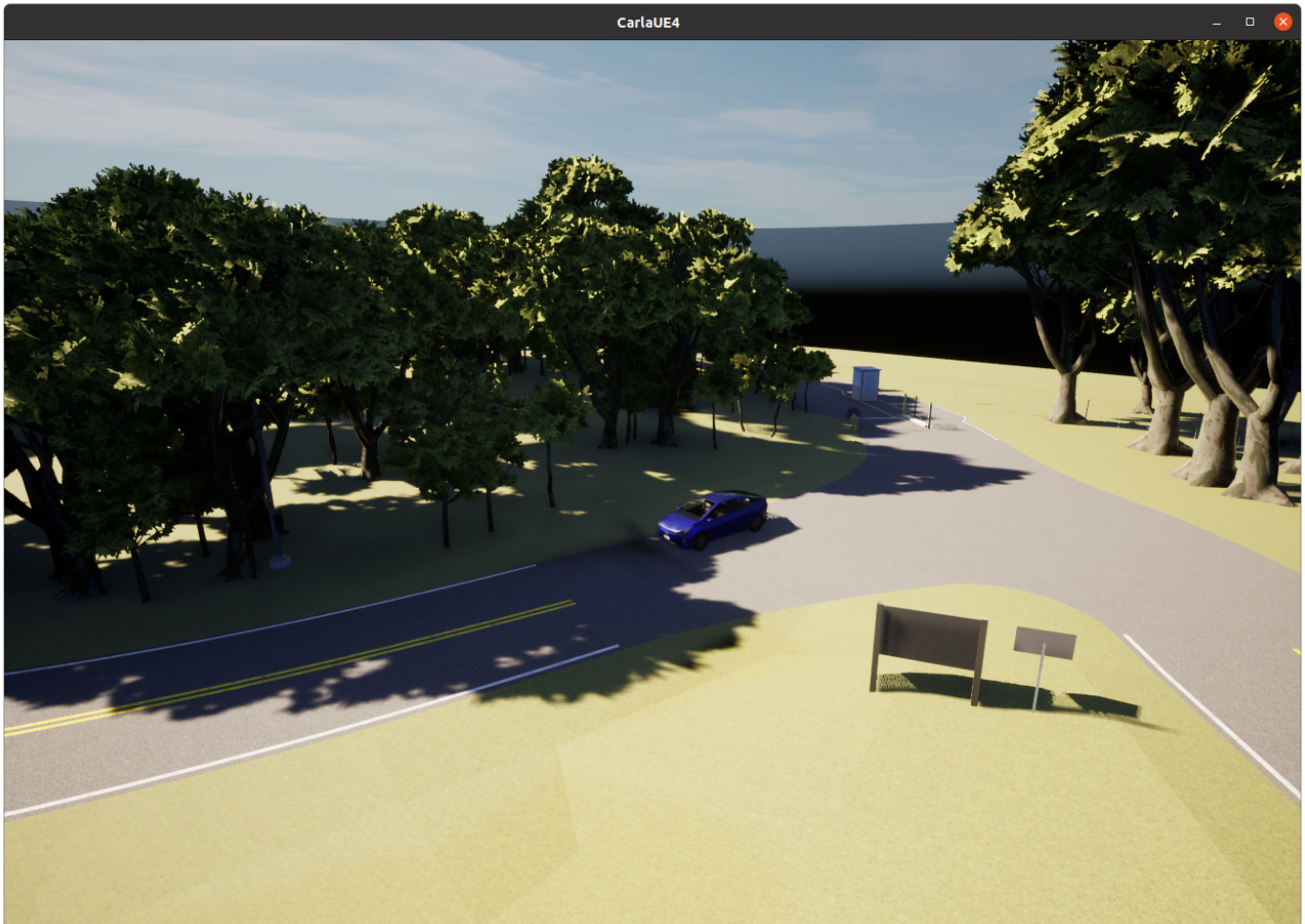
Leave both previously started terminals open and in another terminal, navigate to and start the Carla Simulator. But note, the Carla Simulator and carma-carla-integration tool can time-out, so be prepared to enter the next command as well.

```
cd $HOME/carla/CARLA_0.9.10_TFHRC_Ubuntu_20220301/LinuxNoEditor  
/CarlaUE4 && ./CarlaUE4.sh
```

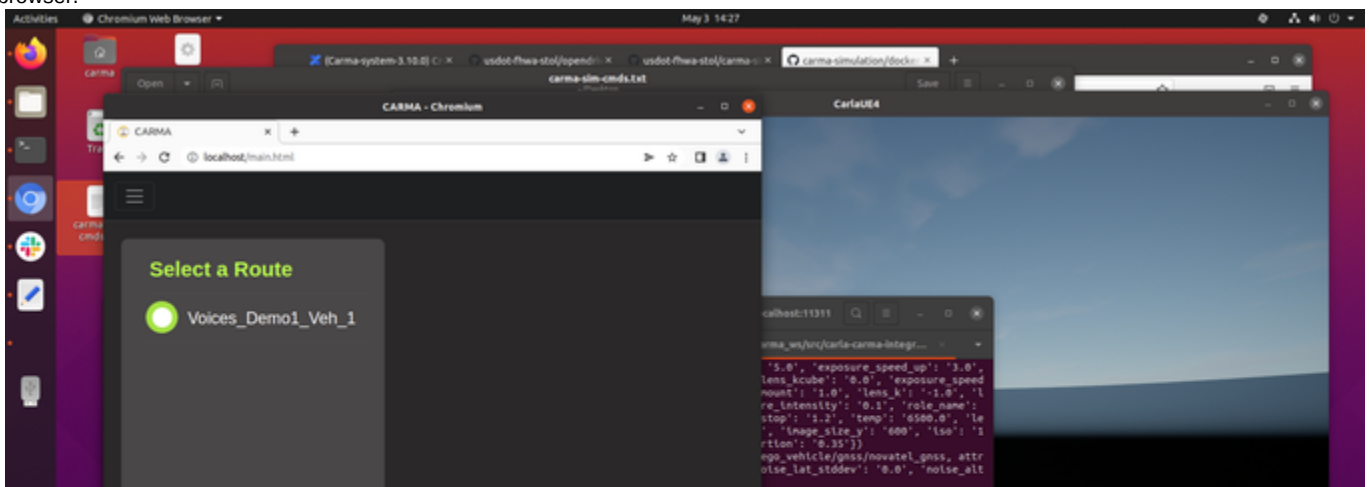
Once the map is loaded, in the original terminal (after running `./run.sh`) start the `carla_carma_agent` and spawn a vehicle with the following command

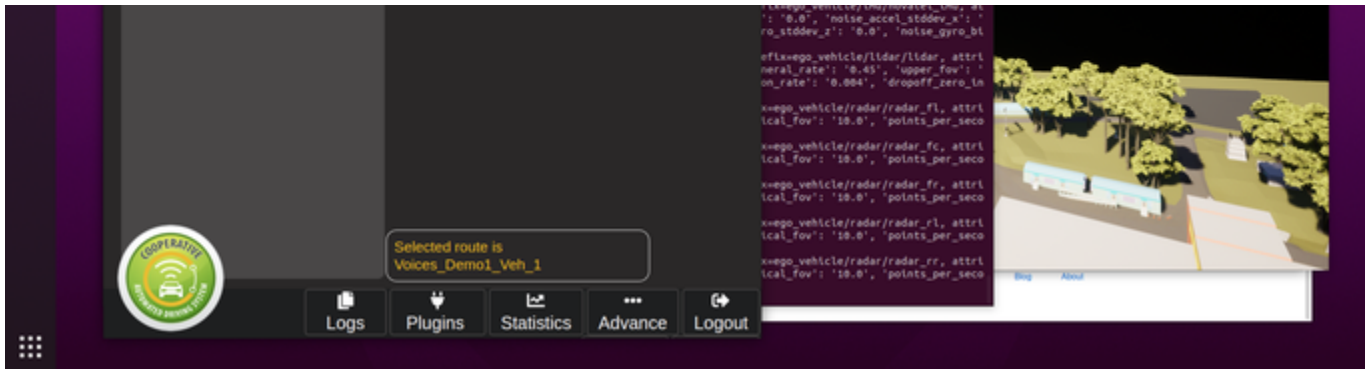
```
roslaunch carla_carma_agent carla_carma_agent.launch town:="TFHRC"
spawn_point:="-174.9, -513.0, 40, 0, 0, 10"
```

You should then see a spawned vehicle in the CARLA simulation



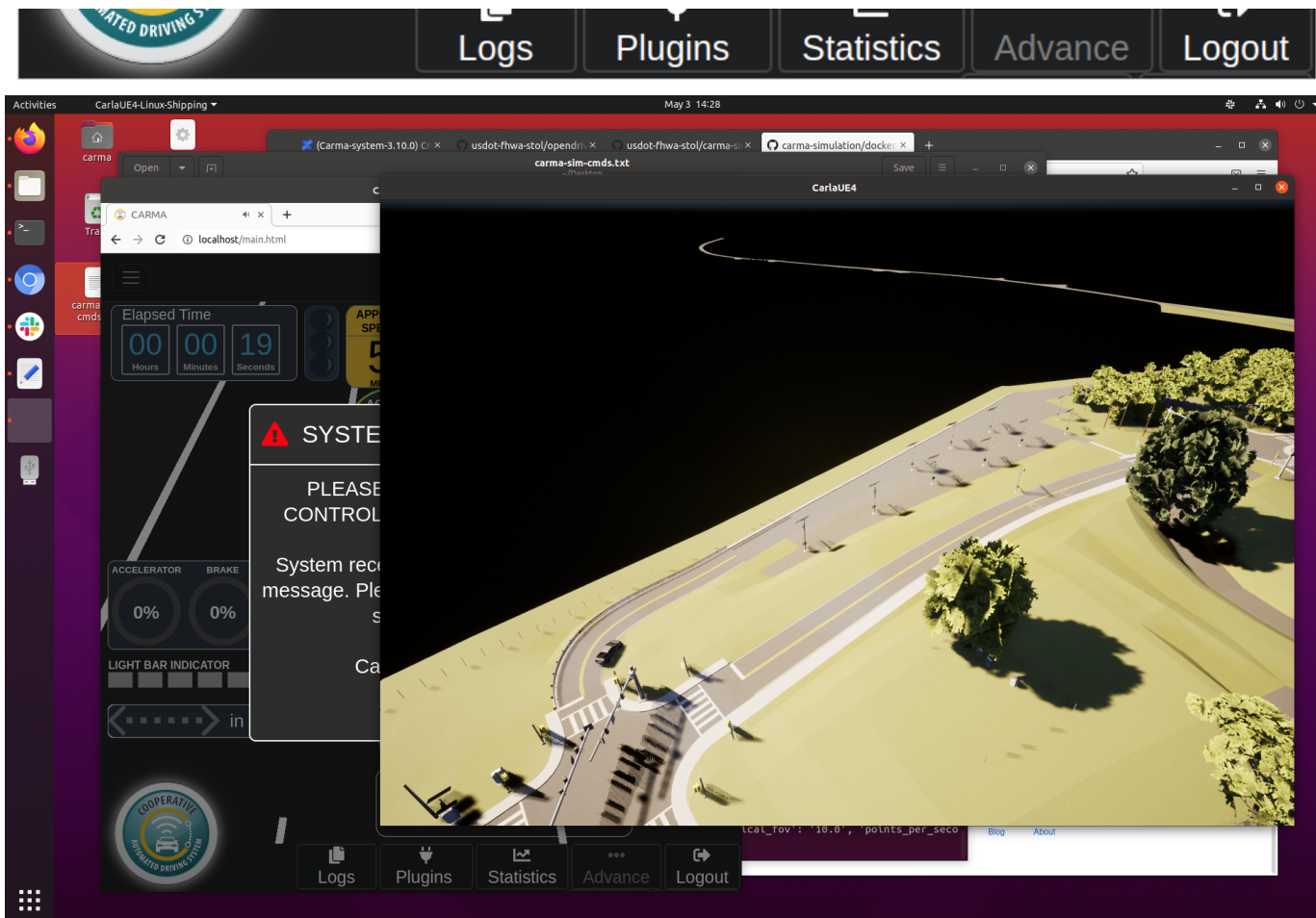
Back in the CARMA Platform Web UI, select `Voices_Demo1_Veh1` Route and click on the green CARMA button in the lower left corner of the browser.





The CARMA Web UI will change to the driving screen and the Vehicle will move along the route previously defined.





Running SIT-1 (previously Demo1)

SIT-1 will consist of 3-4 nodes, all geographically dispersed. Each of the geographic nodes are described below with their location and assets.

McLean, VA - TFHRC

The primary site will be at the TFHRC. It will host the live vehicle, the infrastructure (roadway and traffic lights) as well as the execution manager and other TENA components.

Springfield, VA - DoD Site

A second site, located in Springfield VA will host one CARMA simulation node, including an instance of CARLA and the TENA adapters that are required (carma-platform-tena-adapter and carla-tena-adapter). The sites in Springfield and Augusta are functionally duplicates.

Augusta, GA - DoT Site

A third site, located in Augusta GA will host one CARMA simulation node, including an instance of CARLA and the TENA adapters that are required (carma-platform-tena-adapter and carla-tena-adapter). The sites in Springfield and Augusta are functionally duplicates.

McLean, VA - MITRE

A fourth site, at MITRE in McLean VA will simulate an external (e.g. customer) site. It will host one CARMA simulation node, including an instance of CARLA and the TENA adapters that are required (carma-platform-tena-adapter and carla-tena-adapter). The MITRE site will also be functionally similar to Springfield and Augusta.